



THE ECOLOGICAL AND ENERGY TRANSFORMATION OF UZBEKISTAN WITHIN THE FRAMEWORK OF THE EUROPEAN UNION'S "GREEN FUTURE" INITIATIVE

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Abstract. This thesis explores Uzbekistan's ongoing ecological and energy transformation within the framework of the European Union's "Green Future" initiative. As a strategic partner in Central Asia, Uzbekistan is actively pursuing sustainable development goals by reforming its energy sector, introducing green technologies, and implementing environmental protection policies. The study highlights the country's shift from fossil fuel dependence to renewable energy sources, such as solar and wind power, and discusses the institutional, economic, and international dimensions of this transformation. It also addresses the role of EU cooperation in capacity building, policy development, and investment mobilization.

Keywords: Uzbekistan, European Union, Green Future, renewable energy, energy reform, ecology, sustainable development, green economy

Introduction

Global climate change and environmental degradation have made green transformation a critical priority worldwide. In this context, the European Union (EU) has launched the "Green Future" initiative aimed at supporting its partner



countries in developing sustainable ecological and energy systems. Uzbekistan, as a key partner in the EU-Central Asia cooperation agenda, is actively aligning with this initiative to modernize its economy, transition to clean energy, and protect its natural environment. This article examines Uzbekistan's transformation efforts, institutional reforms, and international collaboration in building a green and sustainable future.

Result and discussion

The EU's "Green Future" Initiative: Overview

The "Green Future" initiative, launched by the European Union (EU), represents a comprehensive framework for fostering ecological resilience, reducing carbon emissions, and advancing renewable energy in partner countries, particularly in Central Asia. The program's main objective is to align global development priorities with the European Green Deal, which envisions achieving full carbon neutrality by 2050 (European Union, 2021). As part of this policy framework, the initiative provides financial assistance, technical expertise, and regulatory support to developing and transition economies such as Uzbekistan, which has only recently embarked on systemic environmental reform.

Environmental and Energy Challenges in Uzbekistan

Uzbekistan faces several interlinked environmental and energy-related challenges that have shaped its developmental trajectory. The country remains heavily dependent on fossil fuels, primarily natural gas and coal, which together account for more than 85% of electricity production (IEA, 2022). This reliance has resulted in high greenhouse gas emissions, exacerbating global warming and contributing to the region's ecological degradation. Another major issue is water scarcity, largely caused by inefficient irrigation systems and the Aral Sea disaster, one of the world's most severe environmental crises (United Nations Development Programme (UNDP, 2023). Compounding these issues are outdated infrastructures within the energy sector, which hinder efficiency and limit the adoption of innovative technologies.



Policy Reforms and Institutional Framework

Recognizing the urgency of environmental reform, Uzbekistan has undertaken substantial policy initiatives over the past decade. The “Strategy for the Transition to a Green Economy (2019–2030)”, adopted by presidential decree, provides a comprehensive roadmap for achieving sustainability across industrial, agricultural, and energy sectors (Government of Uzbekistan, 2019). Complementing this strategy, the National Renewable Energy Development Plan outlines specific measures to diversify energy sources and attract investment in solar and wind power. Moreover, the adoption of the Environmental Code of Uzbekistan (2023) marks a significant step toward integrating environmental governance with national economic planning (Ministry of Ecology, Environmental Protection, and Climate Change, 2023).

These reforms are supported institutionally by the Ministry of Energy and the Ministry of Ecology, which jointly coordinate green policies through public-private partnerships, legislative initiatives, and international cooperation programs.

Renewable Energy Development

Uzbekistan has set ambitious national goals for renewable energy generation: by 2030, renewables are expected to account for 25% of total electricity output, with over 5 gigawatts (GW) of solar and wind capacity installed nationwide (ADB, 2024). Major milestones include the commissioning of large-scale solar power plants in the Navoi and Samarkand regions, as well as wind farm projects in Karakalpakstan. These initiatives have been implemented in partnership with international energy companies such as Masdar (United Arab Emirates) and TotalEnergies (France), which bring both capital and technical expertise (Masdar, 2023). Collectively, these projects are contributing to a measurable reduction in Uzbekistan’s carbon footprint and diversifying its energy portfolio.



EU–Uzbekistan Cooperation in the Green Sector

The European Union has emerged as a strategic partner in Uzbekistan's environmental transformation. Through the European Investment Bank (EIB), the EU provides financial instruments including grants, concessional loans, and investment guarantees to support renewable infrastructure (European Investment Bank, 2022). Additionally, the EU offers technical assistance aimed at strengthening regulatory capacity and training specialists in environmental management. Bilateral engagement is also enhanced by the Enhanced Partnership and Cooperation Agreement (EPCA), which emphasizes joint action on climate resilience and sustainable energy (European External Action Service, 2023).

EU-funded regional initiatives—such as “Sustainable Energy for Central Asia” and “SwitchAsia”—have further contributed to institutional capacity-building, improving the skills of local professionals and encouraging green entrepreneurship in Uzbekistan (European Commission, 2023).

Social and Economic Implications of Green Transformation

The transition to a green economy presents both opportunities and challenges for Uzbekistan. On the positive side, it is expected to stimulate job creation in renewable energy industries, improve public health through reduced air pollution, and strengthen energy independence and national security (World Bank, 2022). At the same time, significant barriers remain, including the high cost of initial investment, the shortage of skilled labor, and institutional resistance from entrenched fossil-fuel stakeholders. To overcome these constraints, the government has launched vocational training programs, public awareness campaigns, and incentive mechanisms to promote green innovation and attract foreign direct investment (UNDP, 2023).

Ultimately, Uzbekistan's environmental transition—facilitated by EU cooperation and domestic reform—signals the country's growing commitment to sustainable development. The synergy between policy modernization, international



partnership, and local capacity-building represents a crucial pathway toward achieving long-term ecological stability and economic diversification. Uzbekistan's ecological and energy transformation, driven by the EU's “Green Future” initiative, reflects a strong commitment to sustainable development and international cooperation. While challenges remain, the progress made in policy reform, renewable energy development, and EU collaboration demonstrates a promising path forward. This transformation not only benefits the environment but also strengthens the country's economic resilience and global standing. Continued support from the European Union will be vital in ensuring long-term success and achieving the shared vision of a greener, cleaner, and more sustainable future for Uzbekistan and the region as a whole.

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YEVROPA ITTIFOQINING “YASHIL KELAJAK” TASHABBUSI DOIRASIDA O‘ZBEKISTONNING EKOLOGIK VA ENERGETIK TRANSFORMATSIYASI

Annotatsiya. Ushbu tezisda Yevropa Ittifoqining “Yashil kelajak” tashabbusi doirasida O‘zbekistonda amalga oshirilayotgan ekologik va energetik



transformatsiya tadqiq etiladi. Markaziy Osiyodagi strategik hamkor sifatida O'zbekiston energetika sohasini isloh qilish, yashil texnologiyalarni joriy etish va atrof-muhitni muhofaza qilish siyosatini amalga oshirish orqali barqaror rivojlanish maqsadlarini faol ravishda ro'yobga chiqarmoqda. Tadqiqotda mamlakatning qazib olinadigan yoqilg'iga qaramlikdan quyosh va shamol energiyasi kabi qayta tiklanadigan energiya manbalariga o'tishi yoritilgan, shuningdek, ushbu transformatsiyaning institutsional, iqtisodiy va xalqaro jihatlari muhokama qilingan. Bundan tashqari, salohiyatni oshirish, siyosatni ishlab chiqish va investitsiyalarni jalb qilishda Yevropa Ittifoqi bilan hamkorlikning o'рни tahlil qilingan.

Kalit so'zlar: O'zbekiston, Yevropa Ittifoqi, Yashil kelajak, qayta tiklanadigan energiya, energetika islohoti, ekologiya, barqaror rivojlanish, yashil iqtisodiyot

ЭКОЛОГИЧЕСКАЯ И ЭНЕРГЕТИЧЕСКАЯ ТРАНСФОРМАЦИЯ УЗБЕКИСТАНА В РАМКАХ ИНИЦИАТИВЫ ЕВРОПЕЙСКОГО СОЮЗА «ЗЕЛЁНОЕ БУДУЩЕЕ»

Аннотация. В данной тезисе исследуется продолжающаяся экологическая и энергетическая трансформация Узбекистана в рамках инициативы Европейского Союза «Зеленое будущее». Будучи стратегическим партнером в Центральной Азии, Узбекистан активно реализует цели устойчивого развития путем реформирования своего энергетического сектора, внедрения зеленых технологий и осуществления политики охраны окружающей среды. В исследовании освещается переход страны от зависимости от ископаемого топлива к возобновляемым источникам энергии, таким как солнечная и ветровая энергия, а также обсуждаются институциональные, экономические и международные аспекты этой



трансформации. Кроме того, рассматривается роль сотрудничества с ЕС в наращивании потенциала, разработке политики и мобилизации инвестиций.

Ключевые слова: Узбекистан, Европейский Союз, Зеленое будущее, возобновляемая энергия, энергетическая реформа, экология, устойчивое развитие, зеленая экономика